1 Attributional Life Cycle Assessment (LCA): LCA aiming to describe the environmentally relevant physical flows to and from a system and its subsystems over their life cycle; Consequential Life cycle Analysis (LCA): LCA aiming to describe how environmentally relevant flows will change in response to poss ible decisions. (Finnveden G, Hauschild MZ, Ekvall T, Guinée J, Heijungs R, Hellweg S, et al. "Recent developments in life cycle assessment". Journal of Environmental Management. 2009;91(1):1-21).

2 Such as for captured CO2 transportation and storage.

fuels/energy carriers, and the TtW emissions (i.e. emissions related to the fuel use) are

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fuels/energy carriers, and the TtW emissions (i.e. emissions related to the fuel use) are quantified regardless of the sourcing/production/conversion/transport and delivery steps of the fuel/energy carrier. WtW emissions are given by the sum of the two par ts, providing the full emission performance associated with the fuel production and use of a certain fuel/energy in a specific converter onboard.

2.4 The GHG emissions are calculated as CO 2-equivalent (CO 2eq), using the Global Warming Potential over a 100 -year time -horizon (GWP100) to convert emissions of other gases than CO 2, as given in the fifth IPCC Assessment Report, 3 for CO 2, CH 4 and N2O, as follows:

2(100) = 2(100) × 2 + 4(100) × 4 + 2(100) ×
2

(CO2 1; CH4 28; N2O 265), this would read as:

2(100) = 1 × 2 + 28 × 4 + 265 × 2)

These GWP100 values should be used for the purpose of quantifying the GHG intensity in accordance with these guidelines.

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SOURCE: MEPC.376(80).pdf
2

(CO2 1; CH4 28; N2O 265), this would read as:

2(100) = 1 × 2 + 28 × 4 + 265 × 2)

These GWP100 values should be used for the purpose of quantifying the GHG intensity in

accordance with these guidelines.

A calculation using a Global Warming Potential over a 20 -year horizon (GWP20) may be provided as information for comparative purposes, as follows:

2(20) = 2(20) × 2 + 4(20) × 4 + 2(20) × 2

(CO2 1; CH4 84; N2O 264), this would read as:

2(20) = 1 × 2 + 84 × 4 + 264 × 2

2.5 These guidelines provide:

.1 WtW GHG emission factors based on a life cycle attributional methodology, expressing the GHG profile of each representative fuel using on Global Warming Potential (GWP) values over a 100 -year time-horizon of included GHG (CO2, CH4 and N2O);

.2 WtT GHG emission factors (CO2, CH4 and N2O) quantified consistently with the attributional approach;

.3 TtW GHG emission factors (CO2, CH4 and N2O); and

.4 sustainability themes/aspects for marine fuels.

2.6 These guidelines define a FLL that carries information about fuel type, feedstock

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SOURCE: MEPC.376(80).pdf
the attributional approach;

.3 TtW GHG emission factors (CO2, CH4 and N2O); and

.4 sustainability themes/aspects for marine fuels.

2.6 These guidelines define a FLL that carries information about fuel type, feedstock used, fuel production pathway, GHG emission factors, information on fuel blends and sustainability themes/aspects.

3 The Global Warming Potential values as given in the IPCC Fifth Assessment Report (AR5) are used in the context of these guidelines.

2.7 The figure below shows a generic WtW supply chain for a fuel. The bunkering marks the last step in the WtT phase before the TtW phase starts.

Figure 1: Generic well-to-wake supply chain

2.8 These guidelines include an initial non-exhaustive list of fuels in appendix 1 depicting the main current and expected future marine fuels.

PART II: METHODOLOGY

3 GENERAL APPROACH

3.1 A Life Cycle Assessment (LCA) based approach provides a holistic assessment of

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the main current and expected future marine fuels.

PART II: METHODOLOGY

3 GENERAL APPROACH

3.1 A Life Cycle Assessment (LCA) based approach provides a holistic assessment of the product/service/system from well -to-wake using data specific to the activity considered. The LCA methodology follows the marine fuel from feedstock sourcing to its utilizati on onboard ships and assesses its life cycle GHG intensity. This approach, applied within the boundaries of the WtW GHG emissions quantification, is applicable across all geographical regions, where emissions occur and allows for quantifying the GHG intensity over the entire fuel/energy supply chain.

3.2 General principles and methodology can be found in ISO 14044:2006 Environmental management Lifecycle assessment Requirements and guidelines . ISO 14040:2006 Environmental management Lifecycle assessment Principles and framework sets the framework for the LCA, for the quantification of the environmental impact of products, processes and services in the supply chain. On this basis, a specific LCA methodology can be tailored for its application to marine fuels.

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framework for the LCA, for the quantification of the environmental impact of products, processes and services in the supply chain. On this basis, a specific LCA methodology can be tailored for its application to marine fuels.

3.3 WtT emissions represent GHG emissions resulting from growing or extracting raw materials, producing and transporting the fuel up to the point of use, including bunkering.

3.4 TtW emissions represent GHG emissions resulting from fuel utilization onboard (e.g. combustion), including potential leaks (fugitive emissions and slip), when relevant for the GHG assessment.

3.5 WtW emissions are the sum of the WtT and TtW emissions and quantify the full life cycle GHG emissions for a given fuel and fuel pathway, used in a given energy converter on board.

3.6 The attributional approach considers all processes along the supply chain of fuel/energy carrier pathways, allowing the quantification of contributions per segment to the

overall GHG intensity of the final fuel/energy product used on board a ship. The expansion of the system boundaries for specific pathways, in which the feedstock or intermediate products

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overall GHG intensity of the final fuel/energy product used on board a ship. The expansion of the system boundaries for specific pathways, in which the feedstock or intermediate products are diverted from existing use(s), may be considered on a case-by-case basis.

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3.7 As regards the expansion of the system boundaries, with consequential elements such as Indirect Land Usage Change (ILUC), concerns with respect to uncertainties and the risk of arbitrariness suggest that the feedstocks with associated ILUC should only be assessed through a risk-based approach, in the framework of sustainability themes/aspects, as part of these guidelines.

3.8 When more than one product results from a conversion process, emissions related to